1. Suppose $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$.
   (a) Compute $P(A \cup B)$
   (b) Compute $P(A^c \cap B)$
   (c) Compute $P((A \cup B)^c)$

2. Let $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \cap B) = 0.1$. Compute $P(A^c \cap B^c)$.

3. Suppose $P(A) = 0.7$, $P(B) = 0.5$, and $P(A \cap B) = 0.35$.
   (a) Compute $P(A \mid B)$
   (b) Compute $P(B \mid A)$

4. Suppose $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.25$. Are A and B independent? Why or why not?

5. A card is drawn at random from a standard 52-card deck. Let A be the event that the card is a heart, and B the event that the card is a face card (J, Q, K). Compute $P(A \mid B)$.

6. A 5-card hand is dealt from a standard 52-card deck. What is the probability that the hand contains exactly 2 hearts? (Express your answer using combinations.)

7. A 5-card hand is dealt from a standard 52-card deck. Given that the hand contains exactly 2 hearts, what is the probability that both of those hearts are face cards?

8. A random variable X has pmf:

$$p(x) = c(x^2 + x), \quad x = 0, 1, 2, 3.$$

   (a) Find c
   (b) Compute $P(X \geq 2)$
   (c) Compute $E[X]$.
   (d) Compute $\mathrm{Var}(X)$.

9. A random variable Y has pmf:

y	1	2	3	4
$p(y)$	0.1	0.2	0.3	0.4

   (a) Compute $P(Y \leq 2)$
   (b) Compute $E[Y]$
   (c) Compute $\text{Var}(Y)$

10. Give an example of two events A and B such that:

   (a) A and B are independent but not mutually exclusive
   (b) A and B are mutually exclusive but not independent

   For each case, describe the sample space and compute $P(A)$, $P(B)$, and $P(A \cap B)$.